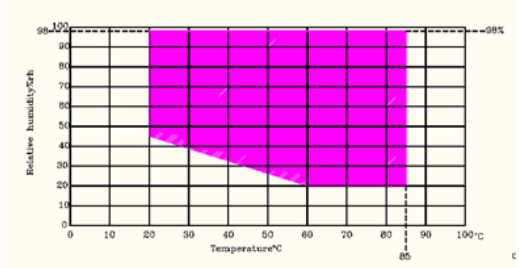
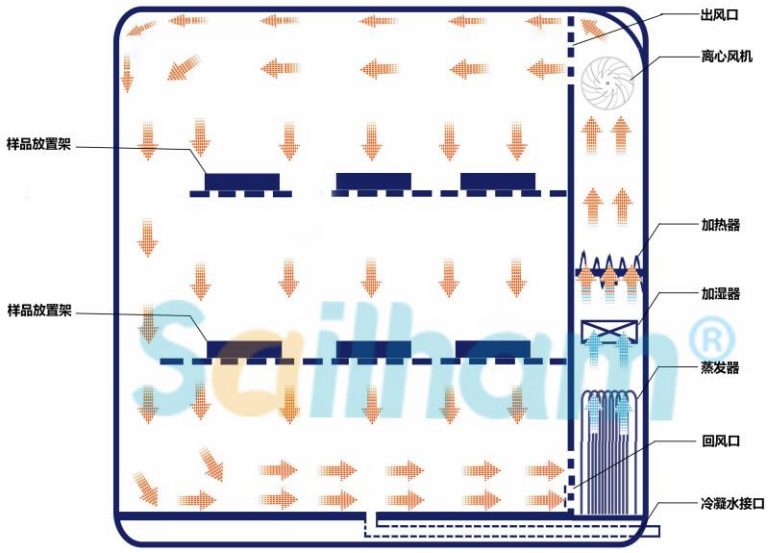


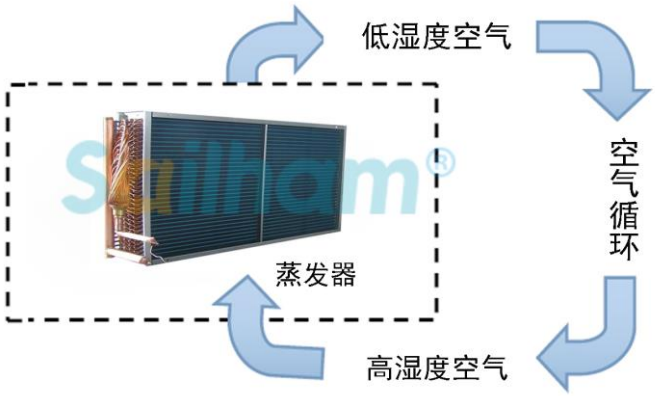
(Reference figure, please refer to the actual product)

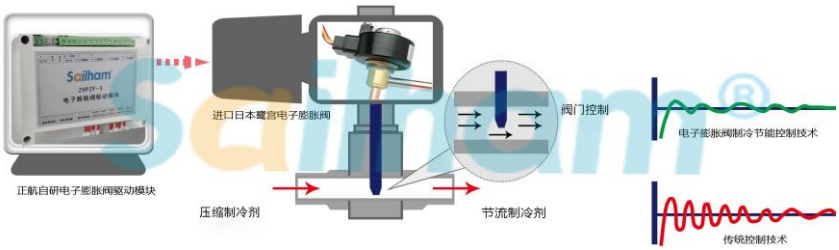
1 Overview	
1.1 Equipment name	Constant temperature and humidity test chamber
1.2 Equipment model	ZH-OHS-408C
1.3 Qualification	Patent certificate number: ZL201620138459. 2 Software certificate number: 2019SR0127335
1.4 Sample restrictions 	● Do not use explosive substances, combustible substances and materials containing the above substances as samples. Otherwise, there is a risk of explosion, fire and other accidents. ● Please do not allow corrosive substances into the tank. If corrosive substances are used in samples or humidification water, especially stainless steel, resin, silicon and other corrosion may lead to a significant decrease in the corrosion resistance of

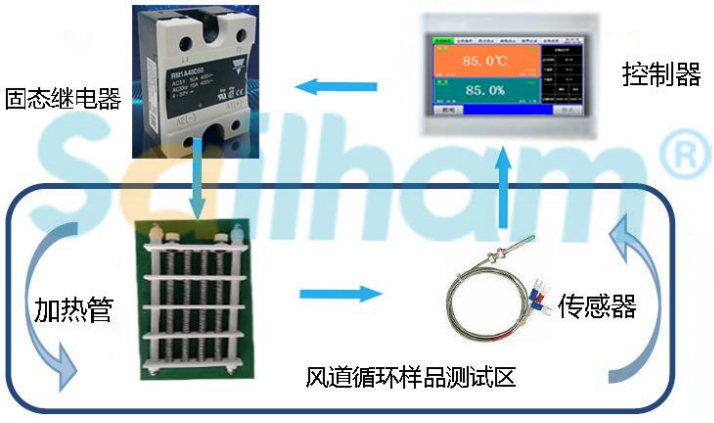
	products. ● Do not use biological materials or substances that exceed the allowable calorific value as samples.
1.5 Equipment purpose	This equipment is designed to evaluate material performance under diverse environmental conditions, including heat resistance, cold resistance, dryness tolerance, and moisture resistance. It serves as a quality assurance tool for testing electronic components, electrical appliances, mobile devices, communication systems, instruments, automotive parts, plastic products, metals, food safety standards, chemical materials, construction materials, medical equipment, and aerospace products.
1.6 Meet the standards	GB/T5170.5 Environmental test equipment inspection of electrical and electronic products GB/10586 Technical conditions of moist heat test chamber; GB T2423.1 Environmental Tests for Electrical and Electronic Products Part 2: Test Methods Test A: Low temperature GB T2423.2 Environmental Tests for Electrical and Electronic Products Part 2: Test Methods Test B: High temperature GB T2423.4 Environmental Tests for Electrical and Electronic Products Part 2: Test Methods test Db alternating wet heat GJB150.3 Environmental test methods for military equipment High temperature test GJB150.4 Military equipment environmental test method: low temperature test.

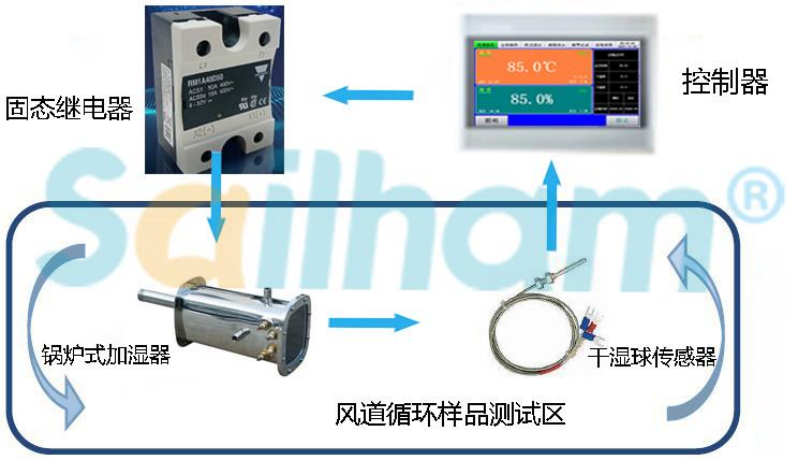
2 size specifications	
2.1 Inner box size	W600*D800*H850mm;容积:408L (liters)
2.2 Outer box size	W890*D1830*H1940mm with the actual object as the standard;
2.3 Sample situation	The sample placement test space is not more than three quarters of the total space, and there is no heat source.
3 equipment performance	
3.1 Temperature range	-20℃～150℃
3.2 Heating rate	≥4.0℃/min nonlinear no-load
3.3 Cooling rate	≥1.2℃/min nonlinear no-load
3.4Temperatureresolution	0.01℃
3.5Temperaturefluctuatin	±0.5℃
3.6Temperature uniformity	±2.0℃
3.7Temperature deviation degree	±2.0℃
3.8 Humidity range	20%~98%R.H (Temperature and humidity control capability is shown below) 
3.9Humidity fluctuation	±2.5%R.H
3.10 Humidity deviation	±3.0%
3.11 Humidity uniformity	±3.0%
* The above temperature and humidity performance is measured under the condition of ambient temperature 23℃±3℃, relative humidity ≤85% no sample unloaded.	
4 Structural features	

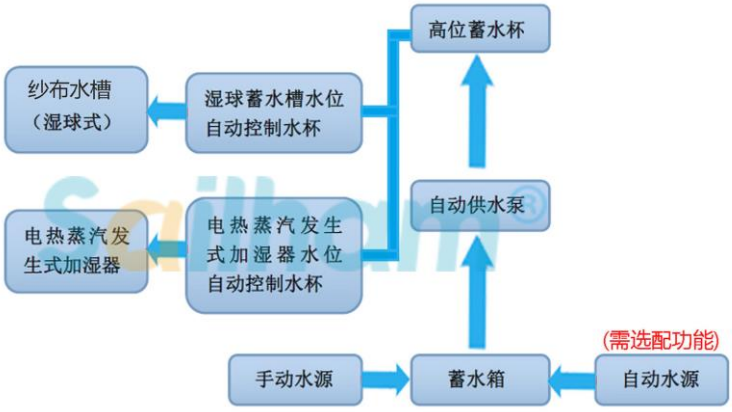
4.1 Inner box material	SUS304 stainless steel 1.0mm.
4.2 Outer box material	Cold rolled steel plate spray paint 1.2mm.
4.3 Insulation material	PU polyurethane high density foam layer, thickness 100mm.
4.4 Air duct circulation system	 The air convection duct circulation system covers the upper outlet air and lower return air, and the temperature and humidity uniformity has a small temperature difference.
4.5 Observation window	The fourth layer is hollow anti-fog glass, and the visible window area is about 250*300mm.
4.6 Wiring test hole	 Φ 1 wiring hole on the left side of the 50mm equipment, equipped with a sealing rubber plug.
4.7 Door lock	 Convenient padlocks
4.8 Control panel	7 inch touch screen controller, power switch, operation indicator light, USB data download interface.
4.9 Lighting	LED high brightness lighting lamp, convenient to observe the

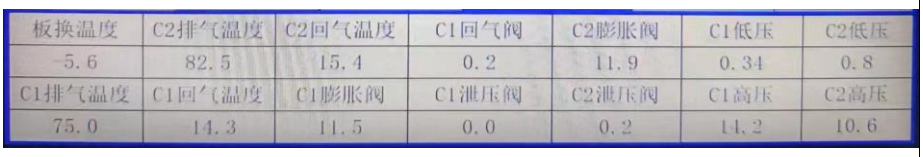
	sample, not blinding.
4.10 Sample shelf	 There are 2 samples shelves with adjustable height of 304 mesh, and each can uniformly carry 20kg.
4.1.1 Power line & Drainage	The rear side of the equipment is equipped with a drainage port and a national standard 2.5m long cable.
5 Refrigeration system	
5.1 Refrigeration working principle (Schematic diagram)	 The refrigeration method of gas compression and conversion uses the physical energy conversion process of refrigerant circulation vaporization and liquefaction in the closed-loop refrigeration system.
5.2 Refrigeration dehumidification principle (Schematic diagram)	 The forced humidity air in the test chamber air circulation system

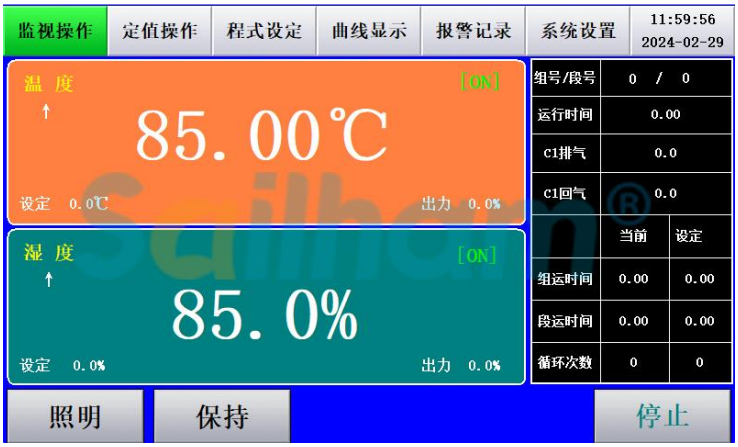
	is extracted from the refrigeration evaporator to absorb water vapor in the air, The air in the test area is dried, and the dehumidification amount is controlled by controlling the refrigeration capacity of the evaporator.
5.3 Energy saving technology advantages of refrigeration electronic expansion valve (Schematic diagram)	 (1) According to different low temperature test conditions, the control system controls the flow opening of the electronic expansion valve through PID algorithm to quickly and accurately control the refrigerant cooling capacity output, so as to ensure that the compressor works in the optimal working output state. (2) More energy-saving than traditional refrigeration technology. Zhenghang's self-developed intelligent control system predicts the low temperature set by users in advance, unloads excess cooling capacity when the low temperature is close to the preset value, and starts heating tube to offset excess cooling capacity, so as to achieve energy saving effect. (3) The energy consumption ratio of refrigeration system is excellently controlled, and the service life of compressor is extended.
5.4 Main refrigeration components	 

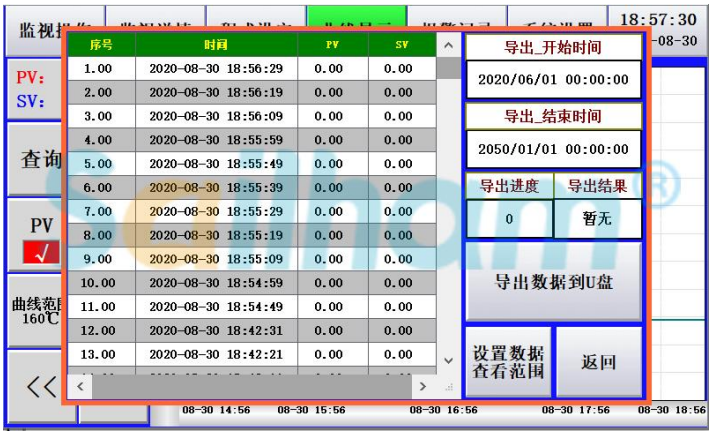
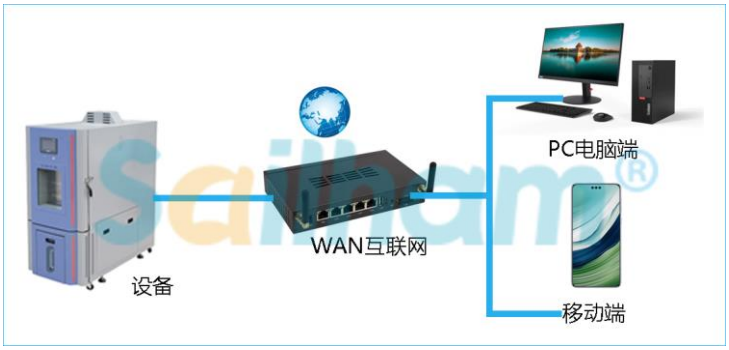
	Air-cooled condenser, fin evaporator, Air-cooled condenser, fin evaporator, Sanyo compressor, Japan
	Sanyo compressor, Japan  American DuPont R404 refrigerant, 
	Japanese Tsurumi pressure sensor,  Japanese Tsurumi electronic expansion valve 
6 Heating system	
6.1 Heating working principle (Schematic diagram)	 The system employs nickel alloy heating wires and P.I.D control algorithms to provide real-time, precise thermal compensation for the chamber's working environment. The closed-loop feedback heating control system utilizes solid-state relays integrated with PID controllers and SSRs as its core components.

	The heaters are strategically positioned within the airflow duct panels, ensuring non-direct radiation exposure to specimens. A modular configuration combined with voltage-regulated heating technology effectively minimizes electrical current surges.
6.2 Temperature sensor	Grade A high precision Pt100 armored platinum resistance is adopted.
6.3 Heating control element	Import Swiss Jiale brand solid state relay, conversion controller output heating control signal.
6.4 Heating devices	Using nickel chromium alloy electric heating wire high temperature resistant ceramic structure.
7. Humidification system	
7.1 Humidification working principle (Schematic diagram)	 The system employs a 304-grade high-strength alloy heating tube with P.I.D control algorithms to provide real-time, precise humidity compensation for the chamber's working environment. The actuation mechanism utilizes solid-state relays with PID+SSR integration. The humidifier is strategically positioned within the ducted airflow plates, ensuring non-direct radiation exposure to specimens. A modular layout design combined with

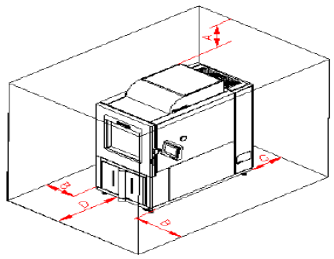
	voltage-regulated humidification technology effectively mitigates electrical current surges.
7.2 Humidity sensor	Grade A high precision Pt100 armored platinum resistance is adopted.
7.3 Humidity control element	Import Swiss Gela brand solid state relay, conversion of the heating control signal output by the central controller.
7.4 Humidification device	High strength alloy heating tube made of 304 material is adopted.
7.5 Humidification and water replenishment system (Schematic diagram)	 Using a wet and dry bulb comparative humidity sensor, automatic water replenishment control is required for the wet bulb sensor (with gauze) and the associated reservoir.
7.6 Requirements for humidification water	Purified water or water with conductivity in the <10us/cm grade must be used. Do not use unfiltered tap water or groundwater as water for the humidification system directly to avoid clogging the waterway and affecting the test effect.
8 control system	

<table>
<tr>
<td>8.1 Control system
advantages
（ self-development ）</td>
<td>(1) The hardware architecture adopts: PLC listed company Xinjie +MCGS (Kunluntong state) 7-inch human-machine interface touch screen.
(2) The software is independently developed by Zhenghang and has the software copyright registration certificate. (It can be customized according to requirements).
(3) Flat UI design, easy to operate, 4GB storage space.
(4) The alarm protection function of high compressor temperature can display the compressor exhaust temperature intuitively on the monitoring screen. (At the same time, it can monitor the working status of components such as the plate temperature and compressor return gas temperature)
 
<table>
<tr><td>板换温度</td><td>C2排气温度</td><td>C2回气温度</td><td>C1回气阀</td><td>C2膨胀阀</td><td>C1低压</td><td>C2低压</td></tr>
<tr><td>5.6</td><td>82.5</td><td>15.4</td><td>0.2</td><td>11.9</td><td>0.34</td><td>0.8</td></tr>
<tr><td>C1排气温度</td><td>C1回气温度</td><td>C1膨胀阀</td><td>C1泄压阀</td><td>C2泄压阀</td><td>C1高压</td><td>C2高压</td></tr>
<tr><td>75.0</td><td>14.3</td><td>11.5</td><td>0.0</td><td>0.2</td><td>14.2</td><td>10.6</td></tr>
</table>
(5) Shutdown State Thermal Protection: When activated, this feature automatically activates the compressor to cool down the system during standby if temperatures exceed the preset over-temperature threshold. Standby Over-Temperature: Defines the minimum temperature at which the compressor will start during standby. Recovery Temperature: Once standby over-temperature activates, cooling will cease when the cabinet temperature drops to or below (Standby Temperature-Recovery Temperature). Delay Activation: Sets the minimum time required before the compressor restarts.
 
<table>
<tr><td rowspan="2">检测有效</td><td>待机超温</td><td>回差温度</td><td>延时生效</td></tr>
<tr><td>155</td><td>5</td><td>60</td></tr>
</table>
</td>
</tr>
<tr>
<td>8.2 Operation mode</td>
<td>● Program: 50 sets of program, each set of program 50 paragraphs, can set all the program cycle, part of the cycle, program connection, set the end action of the program, set the standby action.</td>
</tr>
</table>

	● Setting: the operation time and operating conditions can be set.
8.3 Touch screen	7-inch 800x480 resolution (optional 10,12,15 inches)
8.4 Temperature resolution	0.01℃
8.5 Humidity resolution	0.1%R.H
8.6 Monitoring display function	 Temperature setting SV&PV value, humidity setting SV&PV value, operation time, lighting, heating output power, start operation, compressor exhaust gas return temperature, operation mode and other functions.
8.7 Data storage and functional interface	The human-machine interface touch screen has 4GB of storage space, which can theoretically be stored for about 10 years. (1) USB: You can download historical curves and historical data. The historical data is saved in Excel format, and you can download the required data by date. The curve data can be saved as a screenshot in BMP image format, and you can zoom in, zoom out, and then save the screenshot as needed.

	 (2) RS485 communication interface (provide communication protocol, no software development)  (3) RJ45 network port (communication protocol is provided, software development is not included, LAN local area network connection is supported, WAN external network remote connection is optional)  (Schematic diagram of remote connection)
8.9 Language	Switch between Chinese and English (other languages can be customized)
8.10 Other functions	● Permission password protection can be set ● Fault alarm

	cause prompt ● Power failure self-starting function ● Permission password protection can be set ● Fault alarm cause prompt ● Power failure self-start function ● Power on self-test function ● Upper and lower limit temperature protection function ● Appointment start function ● Power on self-test function ● Upper and lower limit temperature protection function ● Appointment start function
9 Safety protection device	
9.1 Refrigeration system	● Compressor overcurrent protection ● Compressor overpressure protection ● Compressor overtemperature protection
9.2 Electric control system	● Heating and humidification overtemperature protection ● overload, short circuit and leakage protection ● power supply reverse phase protection ● Motor overheating protection ● water shortage protection ● sample over temperature protection function ● Motor overheating protection ● water shortage protection ● sample over temperature protection function
10 Operating environment conditions	
10.1 Equipment placement	(1) The ground is flat and well ventilated (2) No strong vibration around the equipment (3) There is no strong electromagnetic field around the equipment (4) There is no flammable, explosive, corrosive substances and

	dust around the equipment (5) There is appropriate space for use and maintenance around the equipment, as shown in the figure: A: not less than 10cm B: not less than 60cm C: not less than 60cm D: not less than 150cm 
10.2 Environmental conditions	Temperature: +5~30℃ relative humidity: ≤85% air pressure: 86kPa~106kPa
10.3 Power supply	(1) Voltage: AC380V±10%, frequency: 50HZ, protective ground resistance <4Ω. (2) The user shall install the air or power switch with corresponding capacity for the equipment at the installation site, and the switch must be used independently for the equipment (not connected to the socket), and the junction box shall be less than 1.2 meters away from the equipment. (3) Installed power: 10KW maximum current: 18A.
10.4 Humidification water	Purified water or water with conductivity in the <10us/cm grade must be used. Do not use unfiltered tap water and groundwater directly as water for the humidification system to avoid clogging the waterway and affecting the test effect.
11. Appendix list	
Connector hole rubber stopper* 1, sample shelf* 2, operation & maintenance manual* 1, warranty card* 1, wet ball gauze* 1	
12 core configuration lists	

Name of	brand	Brand	Accessory	brand	Brand
navar (software)	Sailham®	China	PCL	XINJE Che	China
compression	Joint	Japan	touch	昆仑通态 kunluntongtai	China
cryogen	DUPONT E.I.	America	solid state	CARLO GAVAZZI Jaya	Switzerland
desiccative	Emerson	America	AC	Schneider Electric Schneider	France
pressure	SAGINOMIYA Egret	Japan	thermal	Schneider Electric Schneider	France
Electronic	SAGINOMIYA Egret	Japan	auxiliary	Schneider Electric Schneider	France
gas-liquid	O&F 欧达●法斯克 Faske	China	Switching	MW MEAN WELL Ming	Taiwan, China
Condensing	MAHLE Mahl	China	circuit	DELIXI ELECTRIC Delixy	China
refrigeration	Polar bears	America	Power on/off	DELIXI ELECTRIC Delixy	China
Sample	Sailham® self-development	China	Phase	DELIXI ELECTRIC Delixy	China
Overtempera	Rainbow rainbow	Korea	Humidifier	Feiyang	China
	Jiahe	China	heating	Feiyang	China
condensator	Jiahe	China	sensor	Min Yang	Taiwan, China
Recirculating	Yi Zheng	China	transformer	Sheng Yuan	China

- Some parts, such as due to the shortage of materials, will be replaced by the same grade brand in order not to affect the delivery time. Please be informed that no further notice will be given!